

Homework 1

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Problem 1. The following table gives the number of commercial airline accidents and fatalities in the United States in the years from 1985 to 2006.

- Find the mean of the number of yearly airline accidents.
- Find the median of the number of yearly airline accidents.
- Find the mode of the number of yearly airline accidents.
- Find the standard deviation of the number of yearly airline accidents.
- Draw a bar chart and a time series plot for the number of accidents per year against year, from 1996 to 2006.
- Draw a histogram for the number of fatalities, from year 1996 to 2006. Use ranges of size 50 to create the histogram, i.e., 0-49, 50-99 and so on.

U.S. Airline Safety, Scheduled Commercial Carriers, 1985–2006

Year	Departures (millions)	Acci- dents	Fatal- ities	Year	Departures (millions)	Acci- dents	Fatal- ities
1985	6.1	4	197	1996	7.9	3	342
1986	6.4	2	5	1997	9.9	3	3
1987	6.6	4	231	1998	10.5	1	1
1988	6.7	3	285	1999	10.9	2	12
1989	6.6	11	278	2000	11.1	2	89
1990	7.8	6	39	2001	10.6	6	531
1991	7.5	4	62	2002	10.3	0	0
1992	7.5	4	33	2003	10.2	2	22
1993	7.7	1	1	2004	10.8	1	13
1994	7.8	4	239	2005	10.9	3	22
1995	8.1	2	166	2006	11.2	2	50

Source: National Transportation Safety Board.

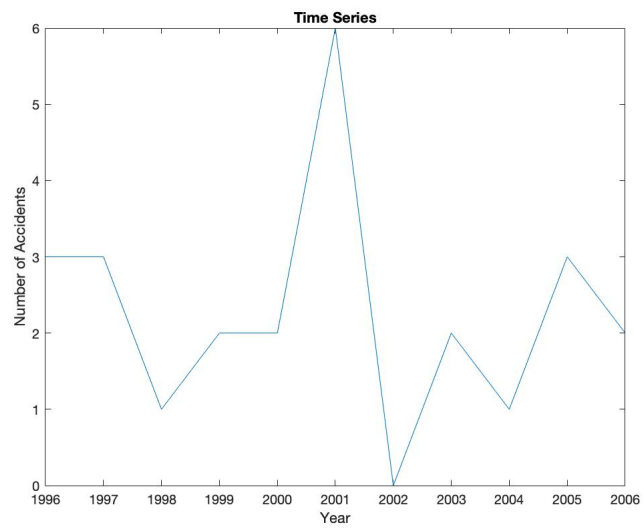
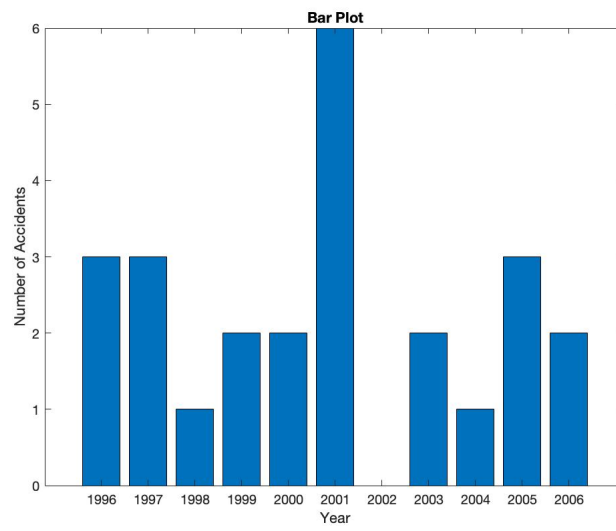
Solution :

There are in total 22 data points, $x_i, 1985 \leq i \leq 2006$, each representing the number of yearly accidents at i th year. For example. $x_{1990} = 6$.

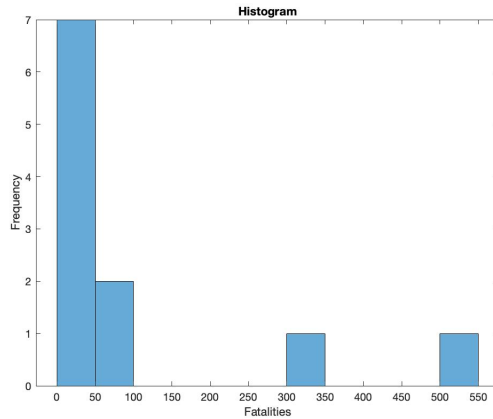
- The mean is $\bar{x} = \frac{1}{22} \sum_{i=1985}^{2006} x_i = 3.18$.
- The median is the average between the 11th and the 12th number after we ranked $x_i, 1985 \leq i \leq 2006$ from the smallest to the largest. The 11th and the 12th number are both 3, so the median is 3.
- The mode is 2 as there are 6 years with number of yearly accidents being 2, which is the most among all data points.

d) The standard deviation is $\sqrt{\frac{1}{22} \sum_{i=1985}^{2006} (x_i - \bar{x})^2} = 2.27$.

e)



f)



Problem 2. A system is composed of four components, each of which is either working or failed. Consider an experiment that consists of observing the status of each component, and let the outcome of the experiment be given by the vector (x_1, x_2, x_3, x_4) where x_i is equal to 1 if component i is working and is equal to 0 if component i is failed.

- How many outcomes are in the sample space of this experiment?
- Suppose that the system will work if components 1 and 2 are both working, or if components 3 and 4 are both working. Specify all the outcomes in the event that the system works.
- Let E be the event that components 1 and 3 are both failed. How many outcomes are contained in event E ?

Solution:

- There are 2 choices (0 and 1) for each component x_i , so the total number of outcomes in the sample space is $2 \times 2 \times 2 \times 2 = 2^4 = 16$.
- $\{(1, 1, 0, 0), (1, 1, 0, 1), (1, 1, 1, 0), (1, 1, 1, 1), (0, 0, 1, 1), (0, 1, 1, 1), (1, 0, 1, 1)\}$.
- There are 2 choices for each of x_2 and x_4 , so there are $2^2 = 4$ number of outcomes in E .

Problem 3. Show that if $E \subset F$, then $\mathbb{P}(E) \leq \mathbb{P}(F)$.

Solution:

Since $E \subset F$, we can let $D = F \cap E^c$. Note E and D are mutually exclusive (since $D \subset E^c$) and that $F = E \cup D$ and

$$\mathbb{P}(F) = \mathbb{P}(E) + \mathbb{P}(D) \geq \mathbb{P}(E).$$

Problem 4. Show that the probability of exactly one of the events E or F occurs is equal to

$$\mathbb{P}(E) + \mathbb{P}(F) - 2\mathbb{P}(E \cap F).$$

Solution: The set where exactly one of the events E or F occurs is $(E \cap F^c) \cup (F \cap E^c)$. Moreover, we notice that $(E \cap F^c)$ and $(F \cap E^c)$ are disjoint since $(E \cap F^c) \subset E$ yet $(F \cap E^c) \subset E^c$. Thus

$$\begin{aligned} \mathbb{P}((E \cap F^c) \cup (F \cap E^c)) &= \mathbb{P}(E \cap F^c) + \mathbb{P}(F \cap E^c) \\ &= (\mathbb{P}(E) - \mathbb{P}(E \cap F)) + (\mathbb{P}(F) - \mathbb{P}(E \cap F)) \\ &= \mathbb{P}(E) + \mathbb{P}(F) - 2\mathbb{P}(E \cap F). \end{aligned}$$

Problem 5. A group of 5 boys and 10 girls is lined up in random order—that is, each of the $15!$ permutations is assumed to be equally likely.

- What is the probability that the person in the 4th position is a boy?
- What about the person in the 12th position?
- What is the probability that a particular boy is in the 3rd position?
- What is the probability that all 5 boys are placed in the first 5 positions of the line?
- What is the probability that all 5 boys are placed in the first 5 positions of the line and a particular boy is in the 3rd position?

Solution

- The probability is $5/(10 + 5) = 1/3$.
- The probability is $5/(10 + 5) = 1/3$.
- A particular boy is equally likely to be in any one of the 15 positions. So the probability is $1/15$.
- The number of permutations for the first 5 positions of boys is $5!$ and the number of permutations for the subsequent 10 positions of girls is $10!$.
Thus the target probability is $\frac{5! \times 10!}{15!} = \frac{1}{3003}$.
- This is similar as the last problem, except the number of permutations for the first 5 positions of boys is $4!$ since we are fixing the 3rd position for a particular boy.
Thus the target probability is $\frac{4! \times 10!}{15!} = \frac{1}{15015}$.

Problem 6. John, Pedro and Rosita each roll one fair die, independently of each other. What is the probability that the score of Rosita is equal to the sum of the scores of John and Pedro?

Solution:

What are the possible outcomes in our event? If we write an outcome as (Rosita's outcome, John's outcome, Pedro's outcome), then all the possibilities are:

$$E = \{(2, 1, 1), \\ (3, 1, 2), (3, 2, 1), \\ (4, 1, 3), (4, 2, 2), (4, 3, 1), \\ (5, 1, 4), (5, 2, 3), (5, 3, 2), (5, 4, 1), \\ (6, 1, 5), (6, 2, 4), (6, 3, 3), (6, 4, 2), (6, 5, 1)\}$$

If we count the number of outcomes in E , we know that $k = 15$. However, the number of possible outcomes for a dice is 6, so if we have 3 dices, the number of possible outcomes would be $n = 6 \times 6 \times 6 = 6^3 = 216$.

$$\text{Thus } \mathbb{P}(E) = \frac{k}{n} = \frac{15}{216} = \frac{5}{72}.$$

Problem 7. You throw a fair die six times in a row. What is the probability that all of the six faces values appear? What is the probability that one or more sixes appear?

Solution:

Again, the probability is given by $\frac{k}{n}$ where k is the number of outcomes in the event and n is the size of all the sample space.

We throw the die 6 times, with 6 possible outcomes each time, hence $n = 6^6$.

The number of possible outcomes that all six values appear is the permutation of the values $\{1, 2, 3, 4, 5, 6\}$, which is equal to:

$$k = P_6^6 = 6! = 720$$

$$\text{So the probability is } P(A) = \frac{720}{6^6} = 0.0154$$

If we call B the event of getting one or more sixes when rolling the dices six times, then B^c is the event that we do not get a six any time we roll the dices.

It is easier to compute $P(B^c)$, as the number of outcomes that 6 does not appear is 5^6 :

$$P(B^c) = \frac{5^6}{6^6}$$
$$\Rightarrow P(B) = 1 - P(B^c) = 1 - \frac{5^6}{6^6} \approx 0.6651$$